

Assignment Week-6 (Full mark 15)

Q1 Kohlrausch's law states that at,

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- (a) Infinite dilution, the equivalent conductivity of an electrolyte is equivalent to the sum of the conductance of the cations and anions.
- (b) Finite dilution, the equivalent conductivity of an electrolyte is equivalent to the sum of the conductance of the cations and anions.
- (c) Both (a) & (b).
- (d) None of the above.

Q2 Identify true and false statements-

1 x 4 = 4

- I. For Strong electrolytes (HCl, KCl etc.), the λ vs $\sqrt{c}$ plot is nearly linear
a) T b) F
- II. The conductance of solution depends on the concentration of the electrolyte's nature.
a) T b) F
- III. For weak electrolytes (CH₃COOH), the λ vs $\sqrt{c}$ plot is straight line.
a) T b) F
- IV. The ionic conductance (λ_o^+ & λ_o^-) can also be determined by transport number.
a) T b) F

Q3. The equivalent conductance of a 0.014(N) Solution of ClCH₂COOH, is 109.0. The ion conductance is 40.2 and 349.8 respectively. Calculate the degree of dissociation and the dissociation constant of ClCH₂COOH.

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a) 0.48, 1.5×10^{-3}

b) 3.5, 1.5×10^{-3}

c) 0.48, 1.5×10^{-3}

d) 3.5, 1.5×10^{-4}

Q4. The velocity at which an ion moves under a potential gradient (1 volt/cm) in a solution is called

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a) Equivalent Conductance.

b) Conductance.

c) Ionic Mobility

d) Degree of dissociation.

Q5. At 18°C , the mobilities of NH_4^+ and Cl^- ions are 6.6×10^{-4} and $57 \times 10^{-4} \text{ cm}^2/\text{volt}/\text{sec}$. Calculate equivalent conductance of ammonium chlorate (NH_4ClO_4) and transport number of cation NH_4^+ .

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a) 8.68 mho cm^2 , 0.464

b) 118.7 mho cm^2 , 0.536

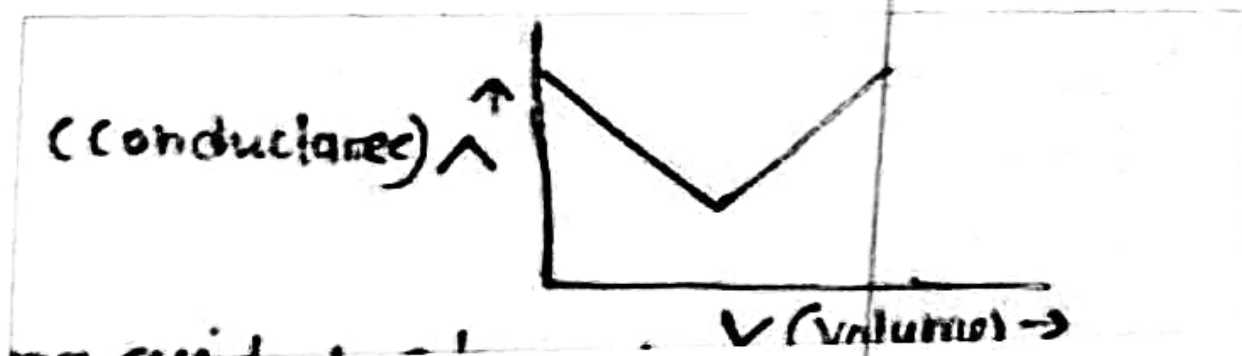
c) 118.7 mho cm^2 , 0.464

d) 8.68 mho cm^2 , 0.536

$$\begin{aligned}\lambda_{eq} &= F [\lambda_+ + \lambda_-] \\ \lambda_{eq} &= 96500 [6.6 \times 10^{-4} + 57 \times 10^{-4}] \\ \lambda_{eq} &= 96500 \times 63.6 \times 10^{-4} \\ \lambda_{eq} &= 61.3 \text{ mho cm}^2 \\ t_+ &= \frac{\lambda_+}{\lambda_+ + \lambda_-} = \frac{6.6 \times 10^{-4}}{6.6 \times 10^{-4} + 57 \times 10^{-4}} \\ t_+ &= \frac{6.6 \times 10^{-4}}{63.6 \times 10^{-4}} \quad t_+ = 0.0104\end{aligned}$$

Q6. The following conductivity titration of

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a) Strong acid + Strong base

- b) Strong acid + Weak base
- c) Weak acid + Strong base
- d) None

Q7. Conductivity of a solution is directly proportional to

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- (a) Dilution
- (b) Number of ions
- (c) Current density
- (d) Volume of the solution